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VU Answer

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Write down the equation of a plane passing through the point $(2, -3, 1)$ and perpendicular to the vector $\hat{n} = 16\hat{i} + 6\hat{j} + 6\hat{k}$.

Answer ([Please click here to Add Answer](#))



State the condition when $\iint_R f(x, y) dA = \int_c^d \int_a^b f(x, y) dx dy = \int_a^b \int_c^d f(x, y) dy dx$ where R is the region of integration.

Answer ([Please click here to Add Answer](#))



Find the critical points for the function. $f(x,y)$ where as $f_x = 2x + y$ and $f_y = 2y + x$.

Answer ([Please click here to Add Answer](#))



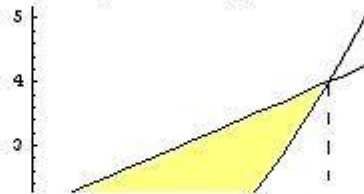
$$\iint (3 + 2x - 3y^2) \, dx \, dy$$

Consider a function $f(x, y) = x^3 - 3xy + y^3$. One of its critical point is $(1, 1)$. Find, whether $f(x, y)$ has relative maxima, relative minima or saddle point at $(1, 1)$.

Answer ([Please click here to Add Answer](#))

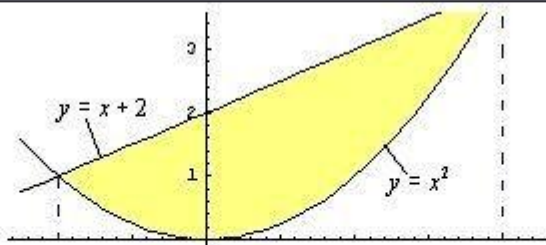


Use double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x^2$ and $y = x + 2$.



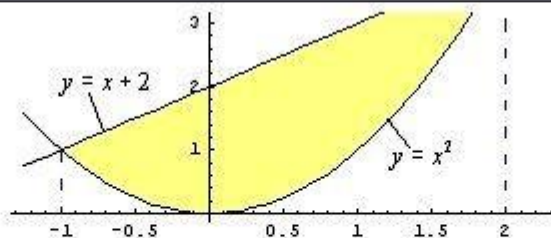
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Rich text editor toolbar showing icons for file operations (New, Open, Save, Print, Copy, Paste, Find), editing (Undo, Redo, Bold, Italic, Underline), and formatting (Text color, Background color, Bulleted list, Numbered list, Indentation, Link, Unlink, Source code). The font is set to Arial, size 12, and the zoom level is 100%.



Answer ([Please click here to Add Answer](#))





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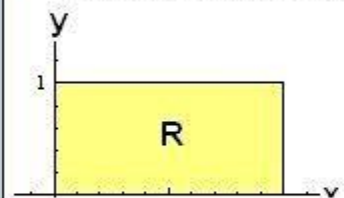
Rich text editor toolbar showing icons for undo, redo, bold, italic, underline, text color, background color, link, unlink, list, and other formatting options. The text "Normal" is selected in the font style dropdown, and "Arial" is selected in the font family dropdown. The font size is set to 12. The zoom level is set to 100%.

Given that $f_{xx} = -6x$, $f_{yy} = -6y$, $f_{xy} = 3$ for some $f(x,y)$. Check whether $f(x,y)$ has relative maximum, relative minimum or a saddle point at P (1,1).

Answer ([Please click here to Add Answer](#))



Let the function $f(x,y)$ is continuous in the region R, where R is a rectangle as shown below.



Answer ([Please click here to Add Answer](#))



Let $y = 3x^2 - 5$. If x changes from 2 to 2.1, find the approximate change in the value of y using differential dy .

Answer ([Please click here to Add Answer](#))



If the order of integration for the integral $\int_0^1 \int_y^1 e^{x^2} dx dy$ is changed. Find the change in the limits of new integral.

Answer ([Please click here to Add Answer](#))



Find equation of normal line (in parametric form) to the surface $f(x, y, z) = xy + 2yz - xz^2 + 10$ at the point $(-5, 5, 1)$.

Answer ([Please click here to Add Answer](#))



Evaluate $\int_1^4 \int_0^2 (6x^2 + 4xy^3) dx dy$

Answer ([Please click here to Add Answer](#))

